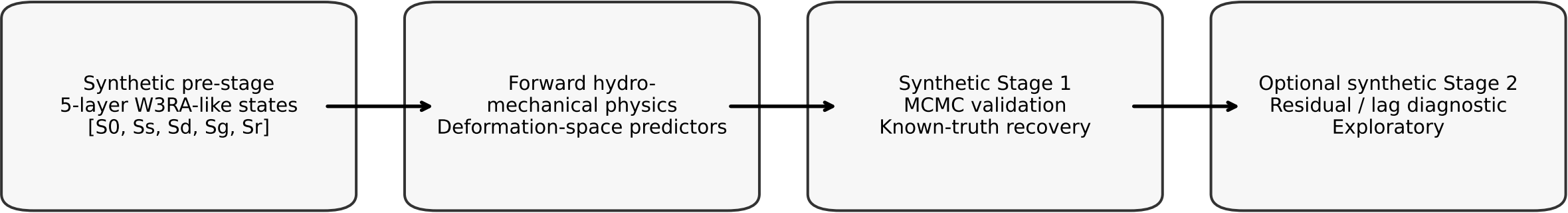
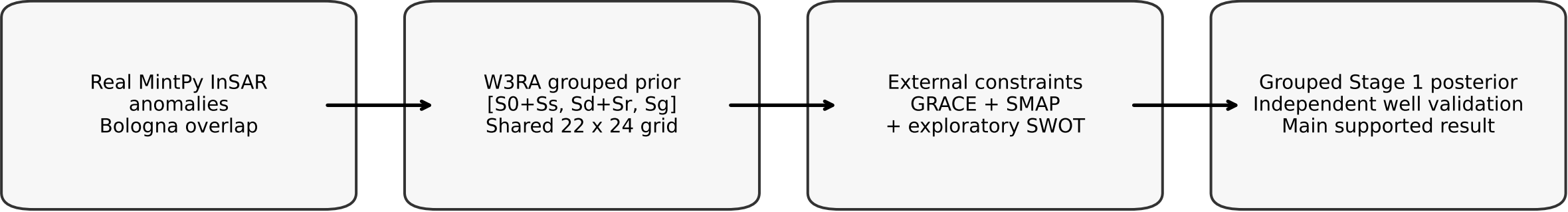


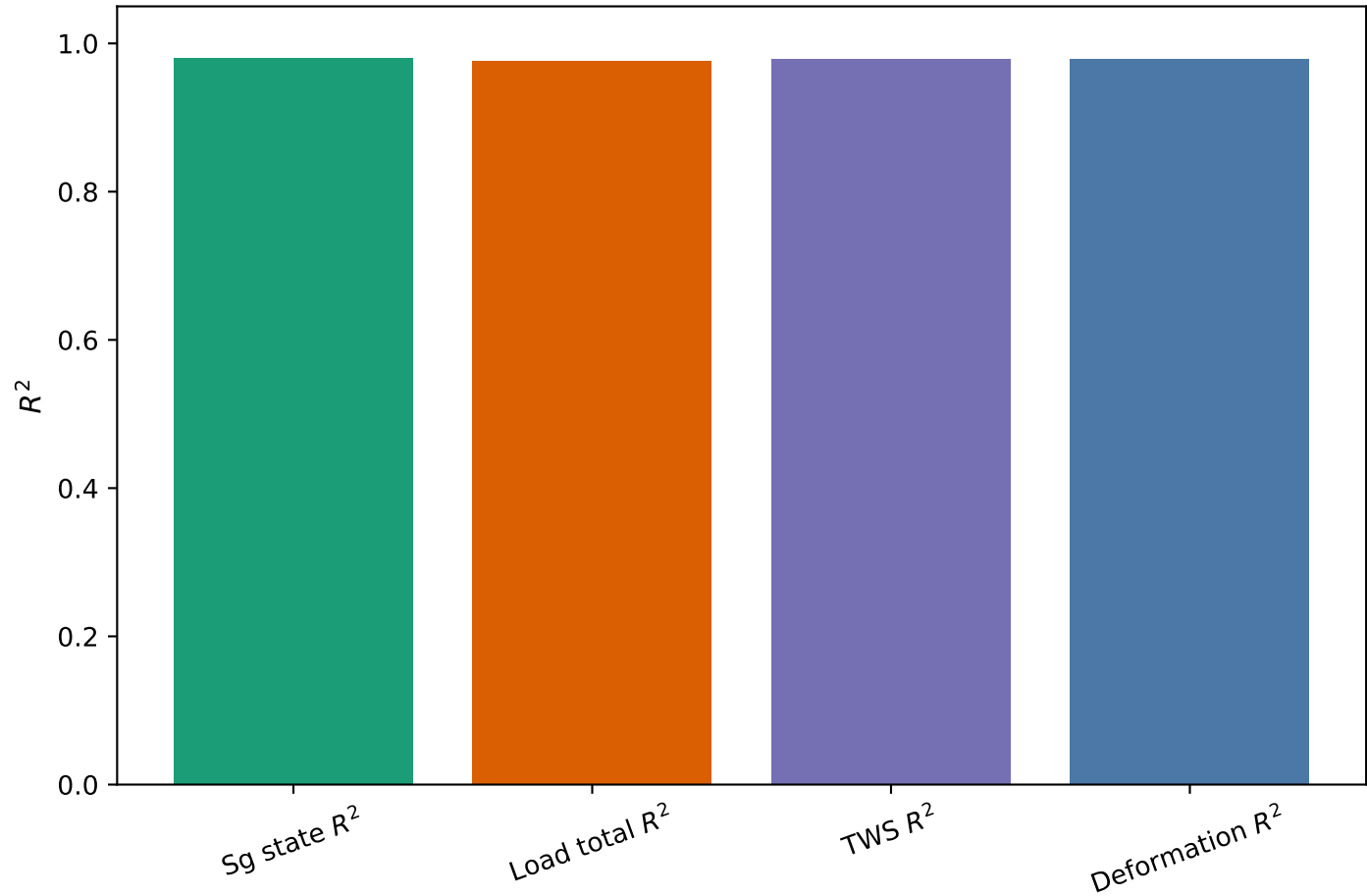
(a) Study workflow



Synthetic pre-stage validates the inversion machinery; the applied Bologna estimator becomes grouped and is externally validated with wells.



(b) Synthetic Stage 1 skill



Grid: 12 x 12
Times: 24
Layers: 5
Noise scale: 0.0

Interpretation:

- Stage 1 recovers grouped synthetic states strongly
- Deformation reconstruction is excellent
- Synthetic success supports the machinery, but does not by itself prove real-data identifiability